Dockerfile Best Practices & Django Template

1. Writing Efficient Dockerfiles

A Dockerfile is a text document that contains all the commands a user could call on the command line to assemble an image. Optimizing your Dockerfile speeds up builds and reduces cloud storage costs.

2. Standard Production Dockerfile for Django

Below is the recommended standard Dockerfile template for deploying Django SaaS microservices:

FROM python:3.11-slim as builder

WORKDIR /app

ENV PYTHONDONTWRITEBYTECODE 1

ENV PYTHONUNBUFFERED 1

RUN apt-get update && apt-get install -y --no-install-recommends build-essential libpq-dev

COPY requirements.txt .

RUN pip install --no-cache-dir --user -r requirements.txt

3. Runtime Multi-Stage Configuration

FROM python:3.11-slim as runner

WORKDIR /app

COPY --from=builder /root/.local /root/.local

COPY . /app

ENV PATH=/root/.local/bin:$PATH

EXPOSE 8000

CMD ['gunicorn', 'config.wsgi:application', '--bind', '0.0.0.0:8000']

4. Key Directives Explained

- **FROM**: Specifies the base image. Use specific tags like 'slim' instead of 'latest' to ensure reproducible builds.

- **COPY**: Copies files from your local computer into the container. Place COPY commands for application code at the very end to maximize cache usage.

- **ENV**: Sets environment variables inside the container environment.